FOUNDATIONS OF GENERATIVE AI (ICT4403)

Assignment 2: Autoregressive Text Generation System
Comparative Study of Simple RNN, Bidirectional RNN, LSTM, and GRU

Student Name	Saksham Gupta
Registration Number	230911186
Assignment Title	Assignment 2 - Autoregressive Text Generation System
System Title	Comparative Word-Level Autoregressive Text Generation on Lewis Carroll's Alice in Wonderland using Simple RNN, Bidirectional RNN, LSTM and GRU
Dataset & Source	Alice's Adventures in Wonderland (Lewis Carroll) https://www.gutenberg.org/files/11/11-0.txt
GitHub Repository	https://github.com/Saksham-Gupta-GH/fgai-autoregressive-text-generation-alice

1. System Overview & Problem Statement

This project implements and evaluates an autoregressive next-word language generation system. Four distinct recurrent neural network architectures were built in PyTorch and evaluated under strictly comparable conditions on Lewis Carroll's *Alice's Adventures in Wonderland*: Simple RNN, Bidirectional RNN, LSTM, and GRU.
To eliminate data leakage, the corpus is split chronologically into 80% Train, 10% Validation, and 10% Test blocks, and the vocabulary is fitted solely on the training tokens (min_freq=2). Input sequences of length 6 predict the 7th token.

2. Quantitative Evaluation Summary (Best Epoch & Test Set)

Architecture	Parameters	Best Ep	Val Loss	Val PPL	Val Top-1	Test Loss	Test Top-1	Test Top-5
Simple RNN	798,484	4	4.0741	58.80	19.73%	4.1584	20.80%	45.28%
BiRNN	1,262,868	2	4.2804	72.27	18.66%	4.3192	18.47%	42.26%
LSTM	1,193,236	5	**3.9903**	**54.07**	**21.87%**	**4.0576**	**23.29%**	**47.23%**
GRU	1,061,652	3	4.0353	56.56	20.55%	4.0900	22.53%	46.51%

3. Cross-Model Temperature Analysis (Averaged Over Seeds)

Temperature	Distinct-1	Distinct-2	Trigram Repetition	Longest Loop	Empirical Quality
T = 0.2	0.517	0.674	23.7%	0–6 words	Grammatically strict; prone to repetitive loops
T = 0.5	0.678	0.900	3.0%	0 words	Optimal empirical trade-off: high diversity with fluent phrasing
T = 1.0	0.855	0.992	0.0%	0 words	High lexical richness; slight syntactic drift across clauses
T = 1.5	0.916	1.000	0.0%	0 words	High entropy; syntactic fragmentation, loss of coherence

4. Conclusion & Model Comparison

• **LSTM edges ahead** across quantitative evaluation metrics: lowest validation loss (3.9903, PPL 54.07), highest validation top-1 accuracy (21.87%), and highest held-out test top-1 accuracy (23.29%) and top-5 accuracy (47.23%).
• **GRU** performs closely behind in second place, achieving 20.55% validation and 22.53% test top-1 accuracy with 11% fewer parameters (1,061,652 vs. 1,193,236). Simple RNN remains competitive (Val Loss 4.0741).
• **Single-Seed Limitation:** Experiments were run under SEED=42 for strict controlled comparison; the 0.045 loss delta between LSTM and GRU falls within typical single-run variance. Multi-seed evaluation (3–5 seeds) is recommended for production benchmarking.
• **Bidirectional RNN:** Concatenating forward final state h_n[-2] and backward final state h_n[-1] feeds complete context to the classifier, but its 1,262,868 parameters overfit by epoch 2 (val loss 4.2804).
• **Sampling Strategy:** T = 0.5 delivers the optimal empirical balance between diversity (Distinct-1: 0.60–0.73) and phrase stability,

maintaining low repetition across models (LSTM 5.0%, GRU 0.0%, BiRNN 1.0%, Simple RNN 6.1%).